ALLEGATIONS AND MIXTURES, SICI

Q1

Directions for question 1: Type in your answer in the input box provided below the question.

Find the average weight (in kg) of a family of five whose weights are 42 kg, 49 kg, 56 kg, 63 kg and 35 kg.

Your Answer: 49

Solution:

The deviation of the 5 weights about an assumed mean of 40 kg are 2, 9, 16, 23 and –5. The net deviation is 45. The actual mean is the assumed mean plus the average deviation, i.e.,

$$40 + \frac{(2 + 9 + 16 + 23 - 5)}{5} = 49$$

Alternately, average $= \frac{42 + 49 + 56 + 63 + 35}{5} = 49$ kg

Ans : (49)

Directions for questions 2 and 3: Select the correct alternative from the given choices.

The simple interest on a sum of money at 4% p.a. for 2 years is Rs.3,750. The sum is _____.

a) Rs.64,875

b) Rs.46,875 (Correct Answer - ✓)

c) Rs.84,675

d) Rs.48,675

Solution:

The SI at % for 2 years is Rs. 3750

$\therefore P\ (2)\left(\frac{4}{100}\right) = 3750$

$\Rightarrow P = \frac{(3750)(25)}{2} = 46,875$ Choice (B)

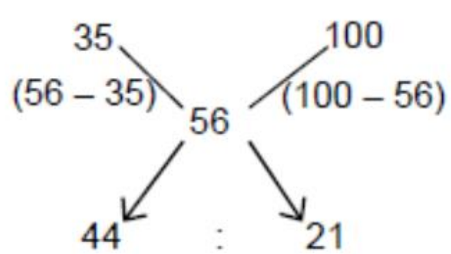
> ***Directions for questions 2 and 3:*** Select the correct alternative from the given choices.

In what ratio must 35% milk solution be mixed with pure milk to get a resultant solution of 56% milk?

- a) 35 : 44
- b) 44 : 21 (Correct Answer - ✅)
- c) 8 : 3
- d) 56 : 9

Solution:

The data is tabulated below.



The required ratio is 44 : 21. Choice (B)

Directions for question 4: Type in your answer in the input box provided below the question.

The ratio of the volumes of milk and water in a solution is 4 : 3. By adding 28 litres of water, the ratio gets reversed and the can becomes full. What is the capacity of the can, in litres?

Your Answer: Unattempted

Correct Answer: 112 ✅

Solution:

The ratio of milk (m) and water (w) is initially 4 : 3 and finally 3 : 4. As the quantity of milk does not change, we express both these ratios in such a way that the number representing the parts of milk is the same. The ratio of m, w is initially 12 : 9 and finally it is 12 : 16 .

∴ If 7 parts of w are added, the ratio gets reversed. As 7 parts is 28 litres, each part is 4 litres. The quantities of m, w are initially 48 and 36. After 28 litres of water are added, the ratio becomes 48 : 64 = 3 : 4. The capacity of the container is 48 + 64 litres, i.e., 112 litres.

Ans : (112)

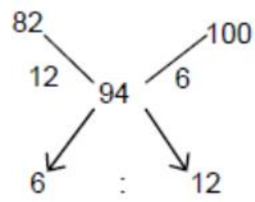
Directions for questions 5 to 7: Select the correct alternative from the given choices.

A sum doubles itself in 6 years and 3 months at simple interest. The rate of interest per annum
is ______.

- **a)** 12%
- **b)** 15%
- **c)** 16%
- **d)** 18%

Solution:

As 6 % water solution means 94% wine,



The required ratio is 6 : 12 or 1 : 2 Choice (D)

Directions for questions 5 to 7: Select the correct alternative from the given choices.

5 kg of ore A contains two-thirds tin and the rest lead. 8 kg of ore B contains one-sixth tin and the rest lead. If equal quantities of the two ores are melted and mixed, what is the ratio of tin and lead in the resultant ore?

- a) 4 : 9
- b) 1 : 2
- c) 14 : 25
- d) 5 : 7

Solution:

The data and calculations are tabulated below, Sn is tin and Pb is lead. The given quantities (5kg of Sn and 8 kg of Pb are not important).

	Sn	Pb	Total	Sn	Pb	Total
A	2	1	3	4	2	6
B	1	5	6	1	5	6
				5	7	

If equal quantities of A and B are mixed, (say 6 parts of A and 6 parts of B), the ratio of Sn and Pb is 4 + 1 : 2 + 5, i.e., 5 : 7. Choice (D)

Directions for questions 8 and 9: Type in your answer in the input box provided below the question.

A certain sum amounts to Rs.6,300 in two years and to Rs.7,875 in three years nine months at simple interest. Find the rate of interest (in %) per annum.

Solution:

The sum amounts to ₹6300 in 2 years and ₹7875 in 3 years 9 months.
Let P be the principal and I be the interest for 1 year

$\therefore P + 2I = 6300,\ P + 3.75I = 7875$

$\therefore 1.75I = 1575 \Rightarrow I = \frac{4}{7}(1575) = 900$

$P = 6300 - 2(900) = 4500$

$\therefore$ The rate of interest $= \frac{900}{4500}(100\%)$ p.a.

= 20% p.a. Ans : (20)

Directions for questions 8 and 9: Type in your answer in the input box provided below the question.

How many kilograms of amla powder costing Rs.42 per kg must be mixed with 21 kg of mehendi powder costing Rs.66 per kg to make a mixture worth Rs.56 per kg?

Your Answer: Unattempted

Correct Answer: 15 ✅

Solution:

The data is tabulated below.

	Amla		Mehendi
Concentration	42	56	66
Quantity			21

The quantities of amla and mehendi are in the ratio (66 – 56) : (56 – 42), i.e. 10 : 14. As 14 parts is 21 kg, 10 parts is 15 kg, i.e., the quantity of amla that has to be mixed with 21 kg of mehendi is 15 kg. Ans : (15)

> ***Directions for questions 10 to 14:*** Select the correct alternative from the given choices.

A sum of Rs.1,200 amounts to Rs.1,800 in ten years under simple interest. What should be the rate of interest so that the same sum amounts to Rs.1,800 in eight years under simple interest?

- **a)** 12.5%
- **b)** 25%
- **c)** 6.25% (Correct Answer - ✅)
- **d)** 18%

🧮 Problem Setup

- Principal = ₹1,200
- Amount after 10 years = ₹1,800
- So, Interest in 10 years = $1{,}800-1{,}200=600$
- Rate of interest for 10 years = $6001{,}200\times100=50\%$
- Therefore, yearly rate = $5010=5\%$

🔎 Step 1: Check condition

We need the rate so that the same sum (₹1,200) amounts to ₹1,800 in **8 years**.

Interest required = ₹600 in 8 years.

🔎 Step 2: Rate calculation

$R=600\times1001200\times8$
$R=600009600=6.25\%$

☑️ Correct Answer

The required rate of interest = **6.25% per annum** (Option c)

💡 Shortcut: When the **same sum** reaches the **same amount** in different times, just use:

$R=100T$

where T= time taken to double. Here, doubling in 10 years → 10%/2 = 5% per year. Adjusting for 8 years gives 6.25%.

> ***Directions for questions 10 to 14:*** Select the correct alternative from the given choices.

Two litres of pure milk is added to six litres of a milk solution containing $16^2/_3$% milk. What is the concentration of milk in the resultant solution?

- a) 12.5%
- b) 25%
- c) $33^1/_3$%
- d) 37.5%

Solution:

$$\text{Final concentration} = \frac{n_1c_1 + n_2c_2}{n_1 + n_2}$$

$$= \frac{2(100\%) + 6(16\frac{2}{3}\%)}{2+6} = \frac{300\%}{8} = 37.5\% \qquad \text{Choice (D)}$$

> ***Directions for questions 10 to 14:*** Select the correct alternative from the given choices.

The average weight of 30 students is x kg and the average weight of 30 other students is y kg. The ratio of x and y is 2 : 3. Find the average weight of the 60 students.

- a) 30 kg
- b) 36 kg
- c) $\frac{2x+3y}{5}$
- d) $\frac{x+y}{2}$

Solution:

The average weight of 30 students is x kg.
The average weights of 30 other students is y kg.
$\therefore$ The average weight of all the 60 students

$$= \frac{30x + 30y}{60} = \frac{x + y}{2}$$ Choice (D)

> ***Directions for questions 10 to 14:*** Select the correct alternative from the given choices.

6, 4 and 6 litres of three different mixtures of petrol and diesel containing 40% diesel, 15% diesel and 4% diesel respectively are mixed. What is the approximate percentage content of diesel in the new mixture?

- a) 19%
- b) 20%
- c) 21%
- d) 22%

Solution:

The quantity of diesel in the mixture is
40% (6) + 15% (4) + 4% (6) = 15% (4) + 44% (6)

$$= \frac{60}{100} + \frac{264}{100} = 3.24$$

The total quantity of the mixture is (6 + 4 + 6) litres.

$$\therefore \text{The content of diesel} = \frac{3.24}{16} = \frac{324}{1600} = \frac{81}{400}$$

$= 20.25\% \approx 20\%$ Choice (B)

Directions for questions 10 to 14: Select the correct alternative from the given choices.

How much will Rs.18,000 amount to in two years at 16% p.a. compound interest?

a) Rs.24,200.8

b) Rs.22,004.8

c) Rs.24,220.8 (Your answer - ✅)

d) Rs.22,400.8

Directions for question 15: Type in your answer in the input box provided below the question.

56 litres of a mixture of milk and water contains milk and water in the ratio 9 : *x*. If four litres of water is added to the mixture, the ratio becomes 3 : 2. Find the value of *x*.

Your Answer: Unattempted

Correct Answer: 5 ✅

Solution:

After the addition of 4 litres of water, the total quantity of the mixture is 60 litres. The ratio of milk and water is 3 : 2. Therefore the quantities of milk and water are 36 liters and 24 liters respectively. Before the addition, they were 36 litres and 20 liters respectively, i.e., the ratio of milk and water was 9: 5, i.e., x = 5. Ans : (5)